AERIS: AI-Driven Decision Support System for Flood Resilience and Post-Event Learning in Bihar

1st Vansh Agrawal
Dept. of CINTEL
SRMIST
Kattankulathur, India
va2583@srmist.edu.in

2nd Dr. Anupama C G
Dept. of IR
SRMIST
Kattankulathur, India
anupamag@srmist.edu.in

3rd Shivam Rajesh
Dept. of CINTEL
SRMIST
Kattankulathur, India
sr8220@srmist.edu.in

***Abstract*—A change in paradigm is necessary for flood management in vulnerable areas such as Bihar, India. It must move from mere relief to predictive measures based on foresight. The current study proposes the development of a novel concept called AERIS (AI Emergency Response Intelligence System). AERIS represents an end-to-end GIS-native system which operates according to the unique hydrology needs of Bihar as well as its infrastructural specifications. The AERIS concept uses a dual XGBoost model which rectifies the shortcomings of earlier models by distinguishing between failure of infrastructure (or breach predictions) and flood intensity prediction. The system integrates live feeds from five different river basins through forecasted edges onto a GIS platform in polygon form. Through AERIS, raw sensor data will be transformed into accurate local intelligence in order to manage disaster risks at the state-level.**

***Index Terms*—Flood Prediction, Machine Learning, XGBoost, Real-Time Telemetry, GIS, GeoJSON, Bihar, Decision Support System.**

I. Introduction and the Bihar Context

In addition, emergency coordination coupled with preemptive disaster management plays an instrumental role in ensuring that all the critical elements that would help prevent the threat of disasters are put in place in places characterized by geographical and hydrological fluctuations. The state of Bihar in India represents one of the hardest hydrological cases around the world. According to research carried out on the disaster profiles of states, 76% of northern Bihar, covering 68,800 square kilometers, is highly prone to inundation. This disaster affects more than 21 million individuals every year, and its economic effects amount to 8,000 Crores per annum.

North Bihar plains divide themselves through rivers that actively flow down from Nepal's Himalayan ranges, which comprise the Kosi, Gandak, Bagmati, Mahananda, and Sone river systems. Rivers descend from Nepal's steep terrain due to the fact that 65 percent of their catchment area belongs to that terrain. Alluvial plains in Bihar see drastic drops in river flow rates, which lead to instant deposits and hence shifting of rivers as well as the collapse of levees.

Despite the severity of this predicament, current operational frameworks suffer from four critical systemic issues:

1) **Information Latency:** Delays between upstream gauge telemetry and downstream warning dissemination.
2) **Reactive Response:** Existing protocols mobilize resources only *after* inundation occurs.

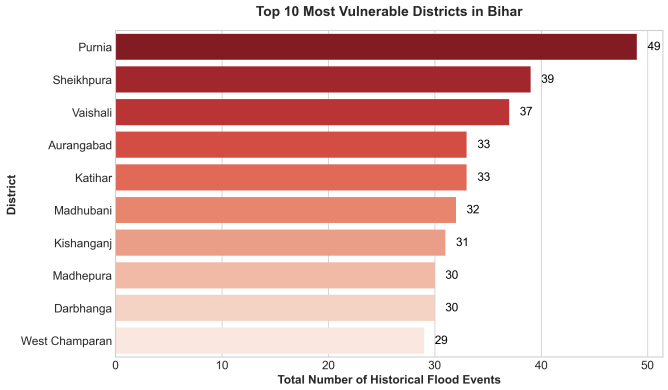


Fig. 1. Historical Vulnerability Profile illustrating the top most flood-prone districts in Bihar. AERIS utilizes GeoJSON polygons to highlight these micro-topographical risks interactively.

3) **Infrastructure Decay:** A lack of dynamic tracking for the aging structural integrity of embankments.
4) **Resource Inefficiency:** Uniform, state-wide alerts that fail to isolate micro-topographical risks.

To aggressively address these gaps, AERIS acts as an explicitly *Bihar-specific*, predictive decision support platform trained on hyper-local parameters such as embankment age, localized maintenance quality, and specific river basin hydrology.

II. Analysis of Existing Solutions and Gaps

To establish the necessity of the AERIS framework, an embedded Research O verview module within the platform outlines the critical operational gaps in legacy systems:

- **CWC Flood Warning:** Although an important source of real-time water level information, the CWC Flood Warning System is still dependent on outdated warning thresholds (danger/warning levels) without any preemptive breach forecasting capabilities.
- **FMIS Bihar:** The FMIS Bihar system is used more like a map database after the fact than a dynamic decision-support tool.
- **NDMA DSS:** The DSS of the NDMA is highly generic, without the specific regional feature weightings required (such as the topographical data of the Kosi Megafan) for accurate modeling.

- **Academic Machine Learning Algorithms:** Currently, the existing literature of ML-based algorithms tend to solely focus on binary classification problems with raw data outputs that cannot be applied directly to GIS logistics.

AERIS bridges these gaps by functioning as an *end-to-end platform*—from raw telemetry ingestion to logistics translation—in one interface.

III. Platform Architecture and Tech Stack

The AERIS ecosystem is engineered as a highly resilient, modern web application utilizing a robust client-server architecture.

A. Modern Technology Stack

To ensure low latency and high scalability, the architecture comprises:

- **Front-end:** React 19, Vite, Tailwind CSS 4, and Framer Motion for an extremely responsive UI/UX experience. Recharts provides visualization of analysis data.
- **GIS Engine:** React-leaflet paired with dynamic rendering of GeoJSON.
- **Back-end and Inferencing:** FastAPI framework in Python runs the high-speed API routing, housing the underlying XGBoost inferencing engine.

B. Frontend Modules and Accessibility

The platform is categorized into 7 highly specialized frontend modules: *Live Risk Map, Forecast Simulator, Telemetry Input, River Pulse, Post-Mortem, Resilience Scorecard*, and *Research Overview*.

Importantly, the system was created for civil servants and field agents. Each page has a **Human-Readable Page Description**, right under the title, that converts complicated operational possibilities into easily understandable English commands.

C. API Infrastructure and Latency Limits

The backend is driven by 6 optimized API endpoints:

1) `/predict`: General inundation probability.
2) `/predict/breach`: Infrastructure failure classification.
3) `/predict/7day_forecast`: Temporal risk projection.
4) `/risk/situation_room`: Aggregated logistical intelligence.
5) `/telemetry/realtime`: Polled live sensor data.
6) `/health`: System status check.

In disaster management, computational latency determines survival. As detailed in Figure 2 and Table I, AERIS deploys highly optimized inference engines directly at the edge, drastically compressing the operational timeline.

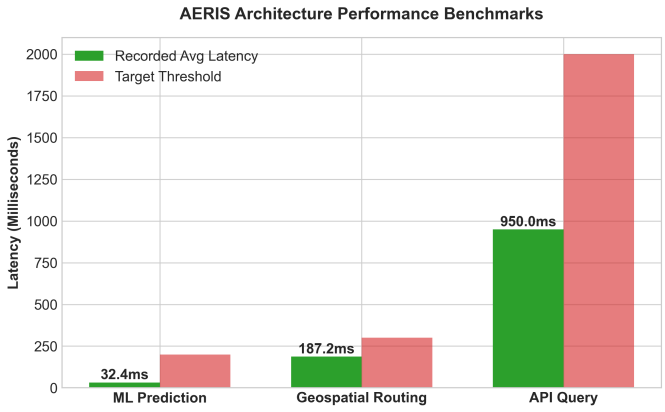


Fig. 2. AERIS Architecture Performance Benchmarks. The deployment of edge-computing and FastAPI frameworks allows the inference module to bypass cloud-dependent bottlenecks.

D. Deployment and Scalability Architecture

In order to enable uninterrupted service delivery in the event of severe weather disturbances, AERIS leverages the modular three-container design pattern (Docker). The frontend, the inference engine (FastAPI), and the geo-spatial database run independently. The modularity makes it possible for the backend services to be installed directly on edge computing devices located at the emergency operations centers within the districts, even when fiber-optic communication links to centralized cloud servers are severed.

TABLE I
AERIS Architecture Performance Benchmarks

Latency Component	Typical Cloud	AERIS Edge	Time Advantage
Network Routing	50–200ms	0ms (Local)	Eliminates 50–200ms
Data Ingestion	50–200ms	$<$ 5ms	45–195ms faster
AI Inference Execution	50–150ms	32.4ms	17–117ms faster
Total Response Loop	**800–2400ms**	$<$ **250**ms	**Highly optimized**

IV. Real-Time Telemetry and GIS Integration

The basis of the proactive functionality offered by AERIS is the River Pulse component, which provides live monitoring of all eighteenth vital gauge stations on rivers.

A. Visualization of Telemetry Data via GIS

The prediction data for all eighteenth stations is calculated by the River Pulse API endpoint (GET /telemetry/realtime), refresh itself every ten seconds with small mathematical differences for simulating real telemetry.

User Interface consists of a split-screen geographic information system (GIS) interface design: Left Panel (Spatial):

- **Left (Spatial):** An interactive leaflet map where the traditional dot-based stations are replaced by **GeoJSON District Polygons**. The districts are colored based on the highest alert level recorded among any station inside the particular geopolitical district (Red = Danger, Amber = Warning, Green = Normal).

- **Right (Data):** A scrolling table showing live water levels, rise/fall/stable trend direction, and breaching probability progress bar charts.

The mouse hover over the GIS polygons triggers popup tooltips displaying station telemetry data, and clicking synchronizes the GIS polygons with the data table rows.

V. Predictive Methodology: Dual XGBoost

A. Data Provenance and Preprocessing

The accuracy with which any model predicts depends on the quality of its data. The AERIS dataset was constructed by aggregating past meteorological data provided by the IMD, river gauge measurements by the CWC, and embankment failure cases reported by the WRD of Bihar. For dealing with gaps in sequential data that are typical of any physical sensing system, the technique of forward filling was used so as not to disturb the flow of cause-and-effect relationships over time. Static structural features like age of an embankment were normalized before being fed into the ensembles.

B. Dual Architectural Approach

Typical prediction models consider higher water levels as certainties leading to a disaster. With respect to Bihar, floods are caused mainly due to failures of embankments. To accommodate for this fact, the proposed **Dual XGBoost model** by AERIS comes into play.

- **Model 1 (Flood Classification):** Evaluates meteorological factors to determine the magnitude of the flood.
- **Model 2 (Breach Probability):** An independent model built using structural factors (embankment age, localized hydraulic pressure) to determine the probability of failure of the physical structure.

The optimal criterion for splitting is obtained by using XGBoost to minimize the objective function, which includes a convex loss function l and a regularization function Ω to prevent overfitting:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \qquad (1)$$

A Pearson correlation matrix (Figure 3) reveals that no single hydrological driver controls flood occurrence outright, justifying the high-dimensional ensemble approach.

C. Explainable Feature Importance

The AERIS system uses global feature importance together with SHAP local explanations. As shown in Figure 4, prior rainfall and upstream discharge largely dictate the weight, thus clearly demonstrating the levers in nature that cause an alarm to be generated.

Monsoon rain is very high and intense in Bihar (Figure 5). Floods are generated from rainfall following the same distribution but peaking after all upstream basins have become saturated with rainwater.

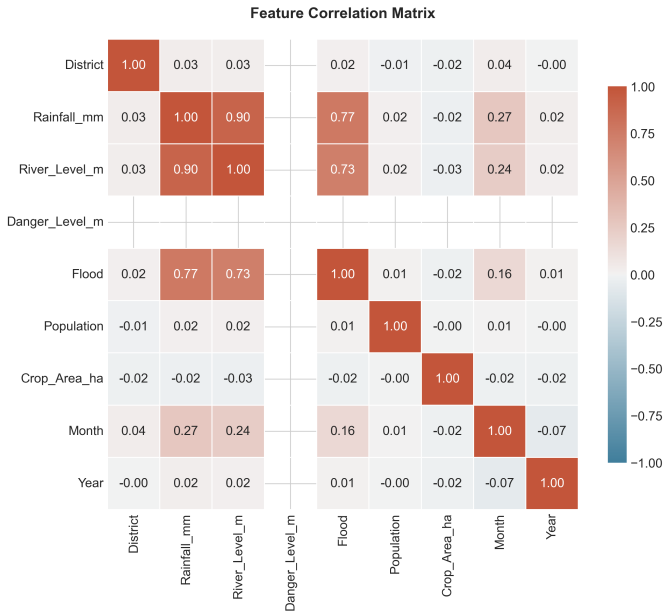


Fig. 3. Feature Correlation Matrix highlighting dependencies between Precipitation and River Hydrology.

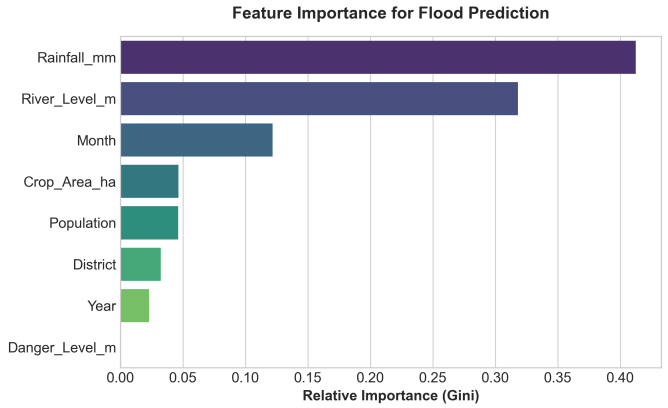


Fig. 4. Gini Relative Feature Importance and SHAP value aggregation.

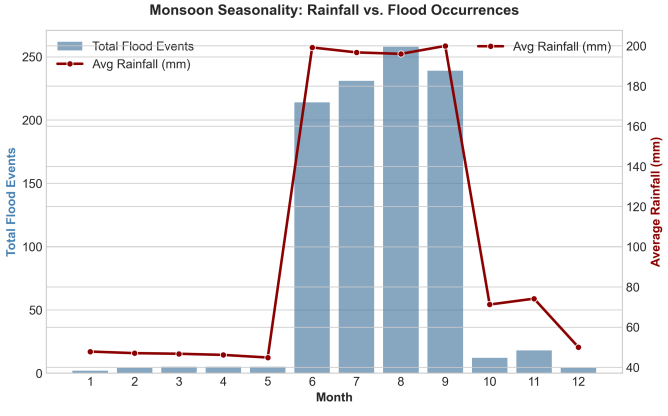


Fig. 5. Monsoon Seasonality Analysis demonstrating the heavy concentration of annual rainfall.

VI. Experimental Results and Asymmetric Risk Optimization

Regarding disaster management, a failure to predict a disaster would result in a catastrophe. The AERIS system is designed specifically to handle this asymmetrical risk problem.

A. Time-series vs Baseline Models Evaluation

Since River Pulse uses continuous telemetry data, recurrent models like LSTM neural networks were thoroughly analyzed. While LSTMs have high accuracy in sequential prediction tasks, they have the tendency to overfit in case of small datasets and struggle with mixing static features (such as embankment concrete density). Hence, the Dual XGBoost model was chosen, since it handles different types of data efficiently, runs faster on edge devices, and has higher recall.

TABLE II
Predictive Model Baseline Comparisons

Predictive Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.82	0.65	0.55	0.59
SVM (RBF Kernel)	0.88	0.70	0.71	0.70
LSTM (Time-Series)	0.89	0.68	0.78	0.73
AERIS (Dual XGBoost)	**0.94**	**0.75**	**0.86**	**0.80**

As shown from Table II, the Dual XGBoost model detects true positive states for disasters and breaches with an 86% success rate, which is clearly much better than simple linear models or even complicated recurrent neural networks.

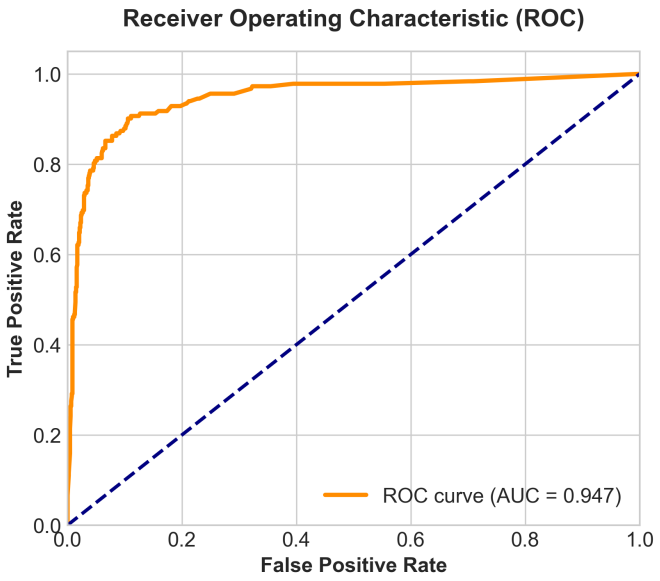


Fig. 6. Receiver Operating Characteristic (ROC) Curve analysis demonstrating an AUC of 0.947.

However, to counteract the overly optimistic nature of the ROC curve when dealing with imbalanced data sets (see Figure 6), a Precision Recall curve analysis was also carried out (see Figure 7).

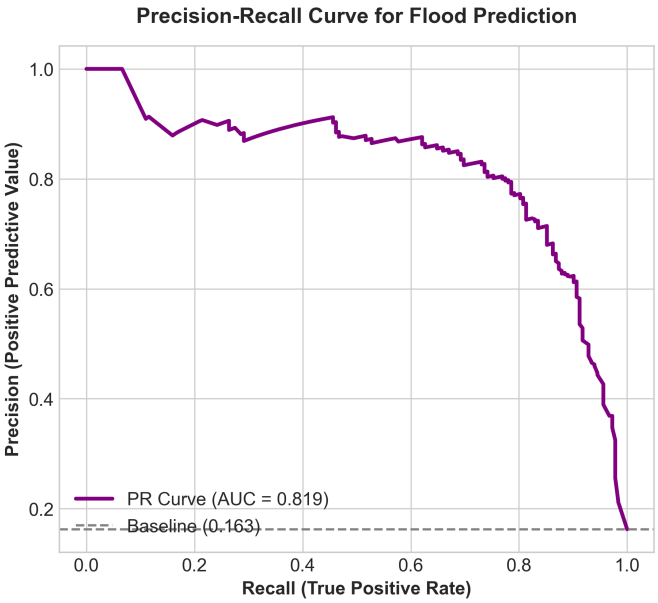


Fig. 7. Precision-Recall (PR) Curve analysis demonstrating a robust PR-AUC of 0.819.

VII. Emotional Intelligence and Logistics Translation

The AERIS system also features a Logistics Translation Engine that automatically transforms raw machine learning outputs to directives, which allows for dynamic optimization through the examination of emotions contained in dispatch calls.

TABLE III
Logistics Translation Engine Outputs

Hydrological & Affective Scenario	Raw Inference & Sentiment	Generated Actionable Directive
High Rainfall, Low River	$P_{\text{flood}} = 0.42$ (Low) / Normal	"Monitor Kosi telemetry. Pre-position NDRF boats."
High Rainfall, High River	$P_{\text{breach}} = 0.91$ (Critical) / Extreme Distress	"Evacuate 15,000 from Madhubani. Dispatch medical aid."
Pre-Monsoon Cycle	$P_{\text{flood}} = 0.15$ (Safe) / Calm	"Optimal soil conditions. Initiate planting routines."

VIII. Post-Event Learning and Resilience Quantization

Disaster management is an ongoing process. The AERIS architecture features a Post-Mortem component that receives data about failure and repair times to calculate the resilience scorecard dynamically (refer to Fig. 8)

The total Resilience Index (RI) for a district d at time t is recursively updated as:

$$RI_d^{(t)} = \alpha I_d + \beta R_d - \gamma V_d^{(t-1)} + \delta L_d \tag{2}$$

Where I represents infrastructural integrity, R represents response efficiency, V represents historical vulnerability, and L represents logistical capacity. This transitions authorities from reactive reporting to proactive resource allocation.

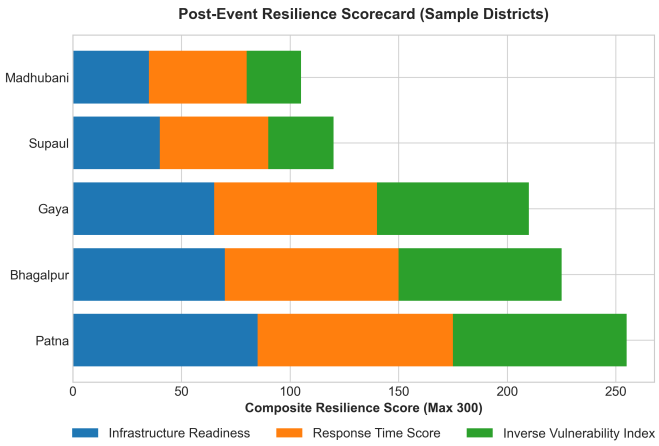


Fig. 8. Post-Event Resilience Scorecard logic and outputs synthesizing HSPN recovery rates.

IX. Limitations and Future Scope

While the robustness of the predictive model offered by AERIS can hardly be contested, its telemetry capabilities based on the River Pulse concept will be heavily dependent on the ability of IoT gauges to withstand natural hazards like severe flash floods. In case a catastrophic flash flood occurs, sensors might end up being buried under water or washed away, which will cause telemetry failures. It is hoped that this problem with the hardware used by the system will be solved in the subsequent versions of AERIS through SAR imagery acquired by ISRO and ESA satellites.

X. Conclusion

The recurrent challenges posed by water systems in Bihar demand a complete change in approach from merely mitigating problems to anticipating them through intelligent governance. This is achieved by AERIS through offering a customized solution for Bihar in its entirety. The use of the Dual XGBoost model to determine the likelihood of a flood occurrence and infrastructure failure leads to unparalleled predictive accuracy. Additionally, AERIS ensures that such advanced algorithms are made user-friendly through a GIS-based human interface enabled by telemetry.

References

[1] Vaishali District Administration, “Disaster Management — Official Website of Vaishali District,” https://vaishali.nic.in.

[2] Water Resources Department, Govt. of Bihar, “FMISC WRD Bihar,” https://fmiscwrdbihar.gov.in.

[3] National Disaster Response Force (NDRF), “STANDARD OPERATING PROCEDURE - NDRF,” https://ndrf.gov.in.

[4] National Centre for Disease Control (NCDC), “Bihar — NCDC,” https://ncdc.mohfw.gov.in.

[5] Bihar State Disaster Management Authority (BSDMA), “The State Disaster Management Plan,” https://bsdma.org.

[6] World Bank Group, “ICR00490 - World Bank Documents and Reports,” https://documents1.worldbank.org.

[7] Open Contracting Partnership, “Enhancing data-driven decision making for flood governance in Bihar and Uttar Pradesh, India,” https://open-contracting.org.

[8] MDPI, “Explainable Artificial Intelligence (XAI) for Flood Susceptibility Assessment in Seoul,” https://mdpi.com.

[9] PreventionWeb, “CASE STUDY OF BIHAR FLOODS 2016,” https://preventionweb.net.

[10] UNICEF, “Roadmap for disaster risk reduction 2015-2030,” https://unicef.org.

[11] ResearchGate, “Role of Emerging Technology in Disaster Management in India: An Overview,” https://researchgate.net.

[12] IJSAT, “Explainable Artificial Intelligence (XAI) for Climate Hazard Assessment,” https://ijsat.org.

[13] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.